

Umma Marium Alam

Full-stack developer, ML researcher, and CS educator — I build things that work, understand why they work, and can explain both.

WEB & APP DEV React · Laravel · MERN · PHP	ML & AI RESEARCH Diffusion · CV · Scikit-learn	TEACHING & MENTORING Python · Web Dev · ML concepts
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ABOUT

I'm a final-year CSE student at BRAC University with a **3.71 CGPA** and a genuinely broad skillset — I've built production-quality web apps in Laravel and MERN, completed and defended a **thesis on diffusion-based AI** with state-of-the-art results, and mentored peers through complex CS topics. I'm currently deepening my ML research through the **GCI World 2026 April program by Japan's Matsuo Lab**. Whether the role calls for building, researching, or teaching — I show up prepared.

SKILLS

FRONTEND & WEB

React.jsJavaScript (ES6+)HTML5 / CSS3Responsive DesignFigma / UI Design

BACKEND & DATABASES

Node.jsExpress.jsLaravel (PHP)MongoDBMySQLREST APIs

MACHINE LEARNING & AI

PythonTensorFlow / KerasScikit-learnDiffusion ModelsComputer VisionPandas / NumPyMatplotlib / Seaborn

TOOLS & WORKFLOW

Git / GitHubJupyter NotebookVS CodeFigmaCanva

LANGUAGES

English (fluent)Bengali (native)

RESEARCH

✓ DEFENDED THESIS · 2025

Fast-DDIM VTON — Multi-View Diffusion-Based Virtual Try-On System
Proposed a novel framework combining Adaptive Cross-Pose Attention, Partial Multi-View Learning, Pose-Adaptive Noise Scheduling, Region-Aware Diffusion, and accelerated DDIM sampling. First system to support front/side/back multi-view synthesis for culturally specific garments — introduced the Bangladeshi Traditional Dress (BTD) dataset. Evaluated on VITON-HD, DressCode, MVG, and BTD benchmarks.

0.8834 0.0987 3.8465 4.1375

SSIM ↑

LPIPS ↓

FID ↓

KID ↓

GCI World 2026 April by Matsuo-Iwasawa Laboratory at the University of Tokyo

2025 · In Progress

[Global AI Research Training](#)

— An online data science and AI course offered by the prestigious Matsuo-Iwasawa Laboratory at the University of Tokyo

ML PROJECTS

Stroke Risk Prediction

2024

[Python](#) · [Scikit-learn](#) · [TensorFlow](#) · [4,000+ patient dataset](#)

- Compared Logistic Regression, Decision Tree, and Neural Network classifiers
- **Logistic Regression: 95.2% accuracy / AUC 0.842** — best overall discrimination
- Analysed class imbalance trade-offs across Accuracy, F1, and AUC-ROC metrics

California Housing Price Prediction

2025

[Python](#) · [Scikit-learn](#) · [Kaggle Dataset](#)

- Linear Regression R^2 0.654 → Random Forest R^2 **0.767** (+17% improvement)
- Applied feature engineering, EDA visualisations, and Grid Search hyperparameter tuning

WEB PROJECTS

Notes App

2025

[MERN Stack](#) — [React.js](#) · [Node.js](#) · [Express.js](#) · [MongoDB](#)

- Full-stack CRUD notes application with React frontend and Express REST API
- Component-based architecture with real-time state updates

Handy Laundry — Online Laundry System

2025

[Laravel](#) · [PHP](#) · [MySQL](#)

- End-to-end booking system with customer flow, order tracking, and admin dashboard
- Role-based access control (customer / staff / admin) via Laravel Auth
- GitHub: [UmmaMariumAlam/Handy-Laundry](#)

Easy Abroad Platform

2024 · In Progress

[React.js](#) · [Node.js](#) · [Express.js](#) · [MongoDB](#)

- Collaborative MERN platform helping students plan study-abroad applications
- University search/filter, application tracker, and document upload — API complete

MovieCard App & Mini Projects (6+)

2024–2025

[React.js](#) · [JavaScript](#) · [Python](#)

- React movie browser with favourites — demonstrates hooks, state, and component reuse
- JS mini projects: slot machine, calculator, hangman, quiz engine, dice game, madlib

EDUCATION

B.Sc. Computer Science & Engineering

CGPA 3.71 / 4.00

BRAC University, Dhaka · Sept 2022 – Present · 12th Semester

- Coursework: Machine Learning, Computer Vision, Data Structures, Algorithms, Software Engineering, Databases
- AWS Educate — Introducing Generative AI with AWS ✓ Completed 2025

HSC & SSC — Science

BNMPC · 2019 – 2021

- GPA 5.00 / 5.00 in both examinations

LEADERSHIP

Director — Peace Café BRACU

2024 – 2026

- Led team operations, event coordination, and member management

Senior Executive — BRAC University Computer Club

2024 – 2026

- Helped in organisation of multiple technical events; mentored junior members in programming and project work

AWARDS



Duke of Edinburgh

Bronze Award · 2024



Data Wizard

Intrahactive 1.0 @ BRACU · 2024



Battle of Insight

Data-Solution 360 · 2024



Runner-up

Sudoku & Creative Writing



Karate

Shotokan 7th Kyu



Agent X Program

2025